

```
from google.colab import files
uploaded = files.upload()
```

[Choose Files](#) 1) iris.csv

1) iris.csv(text/csv) - 3858 bytes, last modified: 5/4/2026 - 100% done
Saving 1) iris.csv to 1) iris (1).csv

```
import pandas as pd
```

```
df = pd.read_csv("1) iris.csv")
df.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species	
0	5.1	3.5	1.4	0.2	setosa	
1	4.9	3.0	1.4	0.2	setosa	
2	4.7	3.2	1.3	0.2	setosa	
3	4.6	3.1	1.5	0.2	setosa	
4	5.0	3.6	1.4	0.2	setosa	

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   sepal_length    150 non-null   float64
1   sepal_width     150 non-null   float64
2   petal_length    150 non-null   float64
3   petal_width     150 non-null   float64
4   species         150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
df.isnull().sum()
```

```

      0
sepal_length  0
sepal_width   0
petal_length  0
petal_width   0
species       0

```

dtype: int64

```
df.describe()
```

	sepal_length	sepal_width	petal_length	petal_width	
count	150.000000	150.000000	150.000000	150.000000	
mean	5.843333	3.054000	3.758667	1.198667	
std	0.828066	0.433594	1.764420	0.763161	
min	4.300000	2.000000	1.000000	0.100000	
25%	5.100000	2.800000	1.600000	0.300000	
50%	5.800000	3.000000	4.350000	1.300000	
75%	6.400000	3.300000	5.100000	1.800000	
max	7.900000	4.400000	6.900000	2.500000	

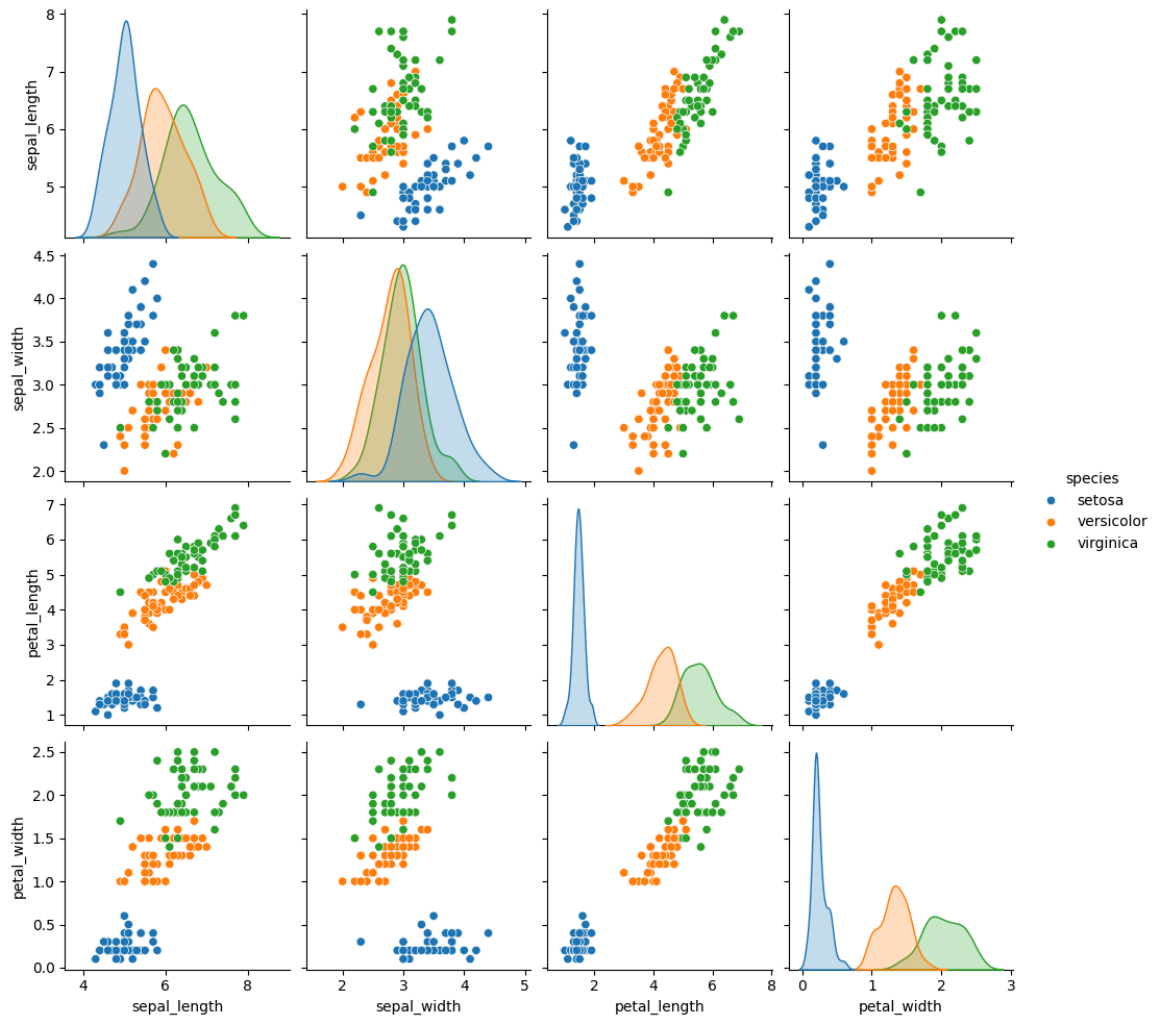
```
df.drop_duplicates(inplace=True)
```

```

import seaborn as sns
import matplotlib.pyplot as plt

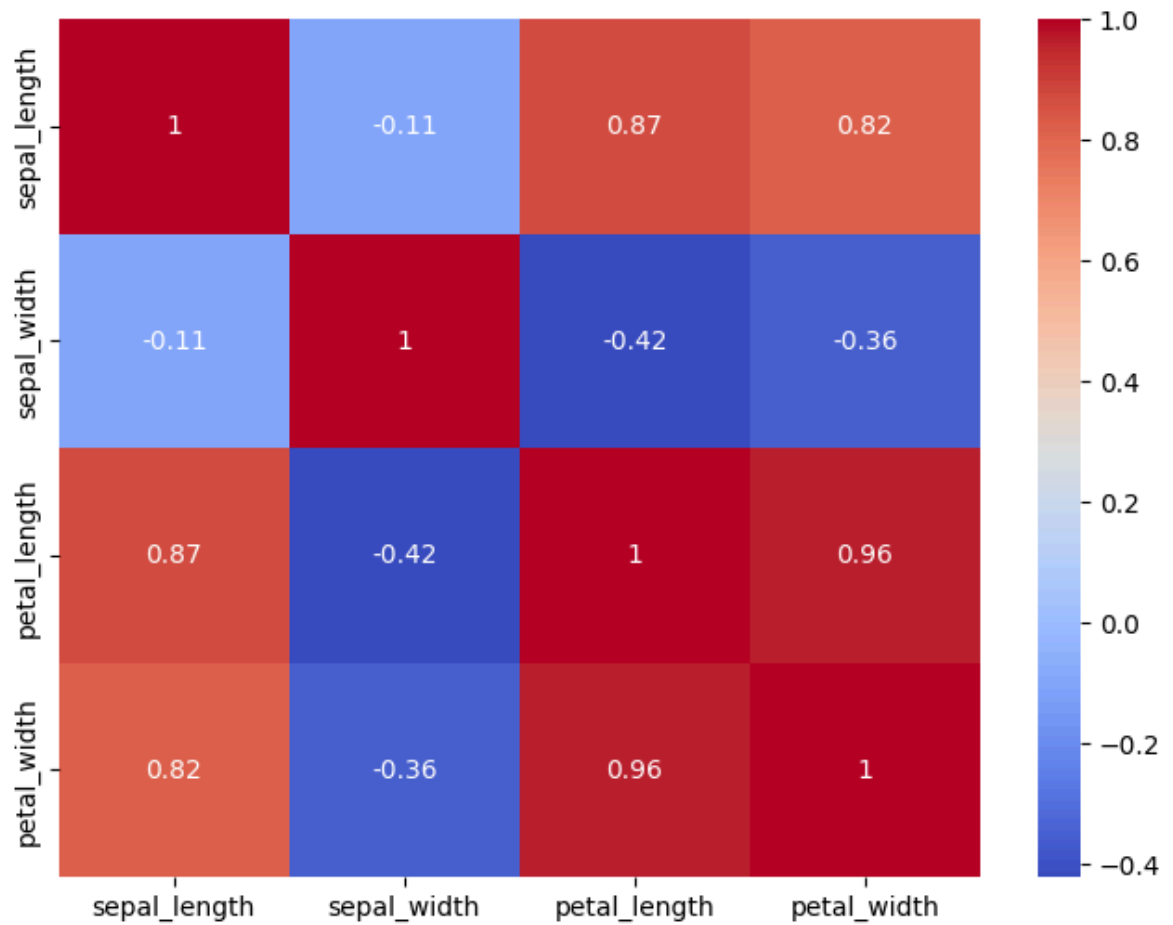
sns.pairplot(df, hue='species')
plt.show()

```

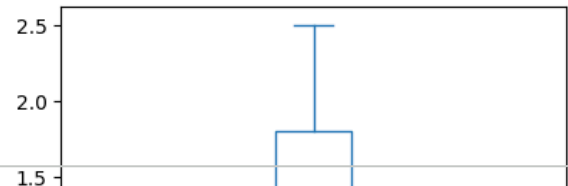
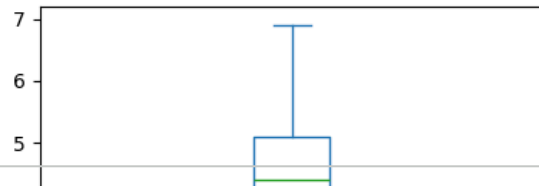
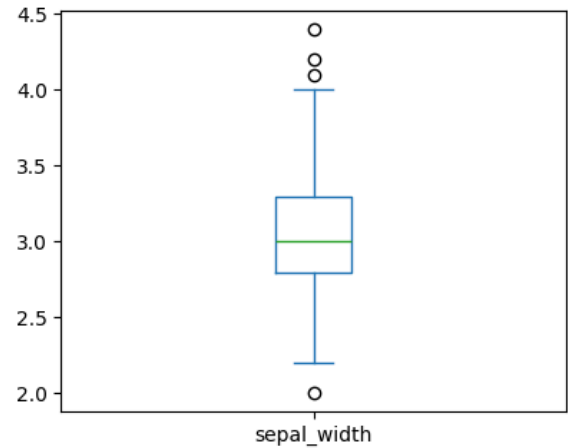
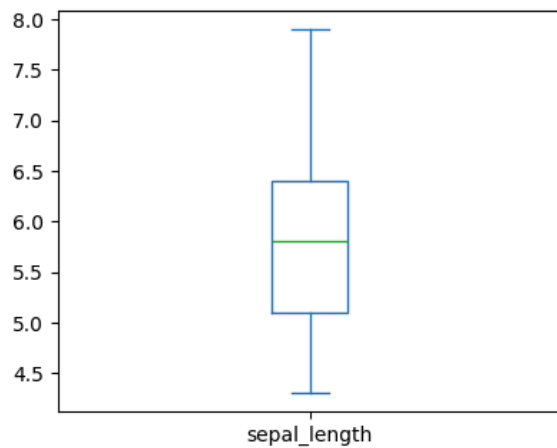


```
numeric_df = df.select_dtypes(include=['float64', 'int64'])

plt.figure(figsize=(8,6))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.show()
```



```
df.plot(kind='box', subplots=True, layout=(2,2), figsize=(10,8))
plt.show()
```



```
df.to_csv("cleaned_iris.csv", index=False)
```

```
from google.colab import files
files.download("cleaned_iris.csv")
```

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

X = df.drop("species", axis=1)
y = df["species"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = RandomForestClassifier()
model.fit(X_train, y_train)

predictions = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, predictions))
```